UNIT I

Foundations of Unsupervised Learning

Discovering useful structure, patterns, or relationships when predefined target answers are not available

Supervised, unsupervised & semi-supervised learning | Problem formulation for unlabeled data | Real-world applications | Distance & similarity measures | Choosing the right measure

"The unit moves from basic learning settings to real-world discovery and finally to technical comparison of observations."

Introduction to Unsupervised Learning

Modern applications continuously generate data from customers, transactions, medical systems, websites, sensors, and social platforms. Much of this information is collected without assigning a predefined category to every observation.

- A customer may not have a known customer segment
- A transaction may not be labeled as normal or fraudulent
- A document may not have a predefined topic
- A biological sample may not belong to a known subtype
- Social interactions may contain communities that were never explicitly defined

The challenge is to discover useful structure from the information already present in the data.

Learning Paradigms in Machine Learning

The learning approach depends on the type of information available and the objective of the problem.

1. **Supervised Learning**
   Learns from examples where the desired target is already known

2. **Unsupervised Learning**
   Works without predefined target labels and searches for useful structure in the observations

3. **Semi-Supervised Learning**
   Combines a limited amount of labeled data with a larger amount of unlabeled data

These paradigms are closely related, but the source of learning information is different: supervised learning relies on known outcomes, unsupervised learning relies only on the structure of the observations, and semi-supervised learning blends both.

Data, Labels, and Learning Objectives

Customer variables	Age, Income, Purchases, Visit Frequency — describe customers but do not say what category they belong to
Now add	Budget / Regular / Premium — a known target

With labels

The problem can be formulated as predicting a known outcome.

Without labels

The problem may focus on discovering groups, relationships, or other structure.

The availability of labels changes the nature of the learning problem: the same customer data can either be used to predict a known outcome or to explore structure that has not yet been defined.

Supervised Learning

Definition: **Supervised learning is a machine-learning approach in which a model learns from labeled examples to predict a target for new observations.**

- Training data contains both features and target values
- The target provides a clear learning objective
- Historical examples guide the learning process
- The model learns relationships between inputs and known outcomes
- New observations receive predicted targets
- Known outcomes allow direct evaluation of predictions

Example: **Predicting loan approval using previous applications, where each historical case already carries a known Approve or Reject outcome the model can learn from.**

Labeled Training Data

Income	Credit Score	Loan Amount	Decision
80K	760	8L	Approve
35K	590	10L	Reject
65K	720	6L	Approve
28K	560	7L	Reject

The first three columns are input features; the final column is the target. During training, the model examines many such examples and learns relationships that can later be used for new applications.

Features and Target Variables

Features

- Age
- Income
- Credit score
- Purchase frequency
- Transaction amount

Target

- Approved / Rejected
- Spam / Legitimate
- Pass / Fail
- House price

Features describe the case; the target defines the outcome the model is trained to predict. Together they form every row of a supervised training set.

How Supervised Learning Works



The model first learns from examples where the correct outcome is available. After training, a new observation is provided without its target, and the model uses the learned relationships to estimate the missing outcome. Because historical targets are known, predictions can be compared with actual results during evaluation.

Classification

Definition: Classification is a supervised-learning task in which the model predicts a predefined category or class for an observation.

- The target consists of categories
- The classes are defined before prediction
- Training examples contain known class labels
- The model learns patterns associated with those classes
- A new observation is assigned to an existing class
- Problems may involve two or several classes

Examples: Spam detection, disease classification, loan approval, fraud classification — in each case the target is one of a fixed, predefined set of categories.

Classification: Spam Filtering

An email service can learn from previously labeled messages: Spam / Legitimate. The content and other measurable properties of the email form the input information, and the previous labels provide the known outcomes.

The model learns patterns associated with unwanted and legitimate messages and applies those patterns to a new email.

Result: **The output is one of the predefined classes. The model is making a prediction, not inventing a new category.**

Regression

Definition: **Regression is a supervised-learning task in which the model predicts a numerical value from one or more input features.**

- The target represents a measurable quantity
- Different observations can have different numerical outputs
- Historical target values guide the learning process
- The model learns relationships between features and the numerical target
- Regression is commonly used for estimation and forecasting

Examples: **House price, monthly sales, electricity demand, temperature — in each case the output is a number rather than a category.**

Regression: House Price

Area	Bedrooms	Age	Price
1200 sq ft	2	8	52L
1800 sq ft	3	5	78L
2400 sq ft	4	3	1.12Cr

Area, bedrooms, and property age are features; Price is the target. A regression model learns relationships from previous properties and uses them to estimate the price of a new property. The output is numerical rather than categorical.

Classification vs. Regression

Aspect	Classification	Regression
Target	Category	Numerical value
Main question	Which class?	What value?
Example	Spam / Legitimate	House price
Typical purpose	Decision	Estimation
Output	Predefined class	Numerical prediction

Both belong to supervised learning because the model learns from examples with known target values. The key difference is the type of target being predicted.



Role and Value of Labeled Data

Labels provide the information that tells a supervised model what outcome it should learn. They allow the system to:

- Define a clear prediction objective
- Learn from historical outcomes
- Compare predictions with actual results
- Measure how well the trained model performs overall
- Support decisions that can be made automatically by systems
- Distinguish between different target classes or values

However, labels are useful only when they are reliable, relevant, and sufficiently available. That requirement creates a major practical challenge.

The Labeling Challenge

Collecting observations is often much easier than assigning reliable labels to them. For example:

- Medical images can be collected automatically
- Expert diagnosis may be required to label those images
- Financial systems can generate millions of transactions
- Investigators may be needed to confirm fraud
- Legal documents can be collected in large volumes
- Trained reviewers may still be required for accurate classification

As data grows, complete manual labeling can become expensive, slow, and difficult to maintain.

Expert-Dependent Annotation

Some datasets require specialized knowledge before a reliable target can be assigned, because a non-expert cannot reliably tell one class from another just by looking at the raw data.

1. **Healthcare**
   A specialist may need to identify a condition from an image
2. **Biology**
   Researchers may need to interpret complex biological measurements
3. **Finance**
   Investigators may need to examine suspicious transactions before assigning a fraud label
4. **Legal data**
   Trained reviewers may be required to classify documents correctly

The difficulty is not only how much data exists, but also how difficult it is to produce trustworthy labels.

Data Volume vs. Label Availability

Data

- Generated continuously
- Often collected automatically
- Can reach millions of observations
- Usually grows faster than manual review capacity

Labels

- Often require human effort
- May require specialist knowledge
- Take additional time to verify
- Can become expensive at large scale

This creates a practical gap: a large amount of available data versus a limited amount of reliable labeled data.

The Label Bottleneck

Total observations	10 million observations
Reliable labels	Only 20,000

The majority of the available information remains unlabeled. Training only on the labeled portion may leave useful patterns undiscovered.

The problem is not that the system lacks data. The problem is that complete supervision is difficult to obtain. This situation motivates learning approaches that can work with data beyond a fully labeled training set.

From Prediction to Discovery

When a target is available, the learning problem can focus on predicting that known outcome. When the target is not predefined, other objectives become possible:

- Find groups of similar observations
- Discover relationships that occur repeatedly within the data
- Identify observations that differ clearly from the rest
- Explore organization that stays hidden within the data
- Investigate patterns not known in advance

The question changes from “What is the expected outcome?” to “What useful structure exists in the data?” This transition leads to unsupervised learning.

Labeled vs. Unlabeled Data

Labeled Data

Observations + Known Targets → Prediction

Unlabeled Data

Observations + No Predefined Targets → Structure Discovery

The observations themselves may remain unchanged. What changes is the information available for learning and therefore the objective of the analysis.



Each point already carries a known label

Unsupervised Learning

Definition: **Unsupervised learning is a machine-learning approach that discovers patterns, structures, or relationships in data without predefined target labels.**

- The data does not contain a supplied target for each observation
- Learning focuses on relationships within the available data
- Groups or other patterns may not be known in advance
- The objective must still be clearly defined
- Results require interpretation to determine their meaning

Example: **Discovering customer groups when no customer segments have been provided — the customers still exist and can still be compared, only the predefined category is missing.**

Unlabeled Data

Unlabeled data contains observations and their available features, but no predefined target that tells the model the expected answer.

A customer record may contain

- Purchase frequency, showing how often the customer buys
- Average spending, showing the typical amount per purchase
- Product preferences, showing which items are favored
- Visit frequency, showing how often the customer returns
- Recent activity, showing the customer's latest behavior

But there may be no column such as Premium, Regular, or Budget. The data is still informative — the missing part is the predefined answer.

Hidden Structure in Unlabeled Data

Definition: **Hidden structure is meaningful organization or relationships that are present in the observations but are not supplied as target labels.**

- Groups of similar observations
- Repeated relationships that occur again and again
- Different behavioral patterns across separate parts of the data
- Unusual observations that stand apart from the rest
- Other forms of organization within the data

The structure is called “hidden” because the system must discover it rather than receive it in advance.

Sources of Hidden Structure

Unlabeled data can still contain useful information through different patterns, even without a supplied target to guide the search.

1. **Similarity**
   Observations may have similar feature values

2. **Relationships**
   Certain items or events may occur together repeatedly

3. **Distribution**
   Most observations may follow a common pattern while a small number differ

4. **Variation**
   Different parts of the dataset may show different behavioral characteristics

These properties provide the basis for discovering structure without predefined labels, because similarity, repetition, and deviation can all be measured directly from the data itself.

Objectives of Unsupervised Learning

The objective of unsupervised learning must be stated clearly because there is no target column defining the expected answer.

1

Group Discovery

Identify similar observations.

2

Relationship Discovery

Identify recurring associations.

3

Anomaly Discovery

Identify observations that differ from dominant patterns.

4

Representation Discovery

Find a useful representation of complex data.

The same dataset can support different objectives depending on the problem being investigated.

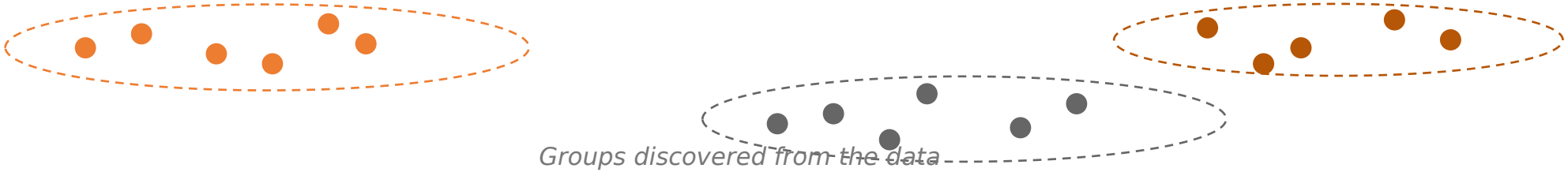
Group Discovery

Group discovery identifies collections of observations that share meaningful characteristics.

Typical situation	A company has many customer records but no predefined customer categories
Objective	Identify groups of customers with similar profiles

The discovered groups may differ in

- Spending, since some groups spend far more than others
- Frequency, since some groups purchase far more often
- Product preferences, showing which items are favored
- Engagement, since some groups interact much more actively



Groups discovered from the data

Relationship Discovery

Relationship discovery looks for items, events, or behaviors that repeatedly occur together.

Example	A retail transaction dataset may contain recurring product combinations, such as Product A ↔ Product B and Product A ↔ Product C

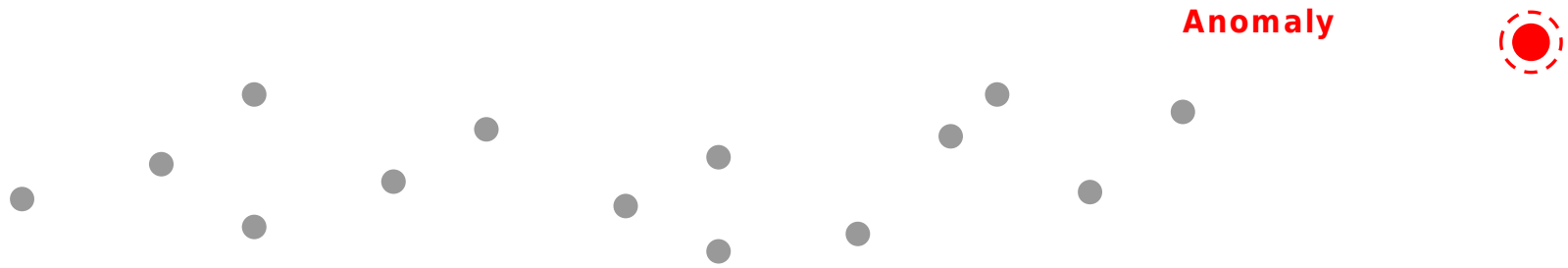
The purpose is not to predict a predefined label. It is to discover relationships that are already present in the observed data.

Anomaly Discovery

Anomaly discovery focuses on observations that differ substantially from the dominant pattern.

Example	In a financial transaction dataset, most transactions may follow a customer's usual behavior while one transaction is very different

Important distinction: **An unusual observation is not automatically an error or fraud. It is simply different from the expected pattern represented by the data.**



One observation deviates from the pattern

Semi-Supervised Learning

Definition: **Semi-supervised learning uses a smaller labeled dataset together with a larger unlabeled dataset.**

This setting is useful when

- Some observations have reliable targets
- Most observations do not have targets
- Complete labeling is expensive or difficult
- The available unlabeled data may still contain useful information

Example: **A medical image collection may contain a small number of expert-labeled scans and a much larger set of unlabeled scans.**

Consider a dataset containing 10,000 observations

Labeled and Unlabeled Data Together

Labeled	500 observations — provide direct target information
Unlabeled	9,500 observations — provide additional observations that may contain useful structure

The learning setting is different from: **Fully supervised (all required targets are available) and fully unsupervised (predefined targets are not available). Semi-supervised means both types of information are present.**

Supervised vs. Unsupervised vs. Semi-Supervised

Aspect	Supervised	Unsupervised	Semi-Supervised
Target labels	Available	Not predefined	Limited
Main objective	Prediction	Structure discovery	Learning with partial supervision
Data	Labeled	Unlabeled	Labeled + unlabeled
Typical output	Predicted target	Discovered structure	Predictions/discoveries using both

Practical distinction: **The main difference is what information is available to guide learning. The choice should also consider what the application is trying to achieve.**

Selecting the Learning Paradigm

Match from: **Semi-Supervised · Supervised · Unsupervised**

1. A retailer has customer records but no predefined customer segments and wants to discover groups.
2. A bank has historical loan applications with known approval decisions and wants to predict decisions for new applications.
3. A hospital has a small labeled image collection and a much larger unlabeled collection.

Answer key: **1 → UnSupervised 2 → Supervised 3 → Semi-Supervised**

The correct choice depends on the available information and the intended learning task, not on which method sounds more advanced — the paradigm should match the data actually on hand.

Problem Formulation for Unsupervised Learning

An unlabeled dataset alone does not fully define an unsupervised-learning problem. A useful formulation must specify:

- What one observation represents
- Which features describe that observation
- What structure should be discovered
- What form the output should take
- Why the result is useful

"Find patterns in this data."

The goal is to convert this broad statement into a specific and meaningful learning problem.

Business statement: “Understand different types of customers.”

From Real-World Problem to ML Problem

Observation	Customer
Features	Spending, frequency, recency, product behavior
Target	No predefined segment
Objective	Discover meaningful customer groups

The technical problem becomes clearer once the original business goal is translated into specific data and discovery terms.

Observation Unit

Definition: **An observation unit identifies what one record represents. It is the basic unit of comparison, and every later step of the analysis inherits this choice.**

- One customer, when analyzing overall customer behavior
- One transaction, when analyzing individual purchase events
- One patient, when analyzing medical or health records
- One document, when analyzing text or written content
- One biological sample, when analyzing lab or genetic data
- One social-network user, when analyzing online interactions

The choice matters because it determines what the learning system is comparing. Grouping customers is different from grouping transactions, even when both come from the same organization.

Feature Representation

After defining the observation, we need to decide how it will be represented, because the model can only compare what is explicitly captured as a feature.

For a customer, the representation could include

- Purchase frequency, showing how often the customer buys
- Average spending, showing the typical amount per purchase
- Recency, showing how recently the customer last purchased
- Product diversity, showing the range of items bought
- Visit frequency, showing how often the customer returns

The representation determines which characteristics are available for comparison. A meaningful representation can reveal useful structure, while irrelevant features can make the discovery less useful.

Relevant vs. Irrelevant Features

Relevant feature	Irrelevant feature
A variable that helps describe the property related to the discovery objective.	***A variable that adds little useful information or introduces noise.***

Useful	Spending, purchase frequency, recency
Possibly irrelevant	An unrelated internal record number

Careful feature selection helps the learning method focus on the structure that actually matters.

Representation Changes the Discoverable Structure

Representation A: Age + Income

May reveal groups based mainly on demographic and financial characteristics.

Representation B: Spending + Frequency + Recency

May reveal groups based mainly on purchasing behavior.

The underlying customers have not changed — the information used to compare them has changed. The discovered structure depends strongly on how the observations are represented.

Discovery Objective and Expected Output

Objective	Expected Output
Group discovery	Groups or segments
Relationship discovery	Recurring associations
Anomaly discovery	Unusual observations that stand apart from the rest
Representation discovery	A useful compact representation

A clear output helps connect the machine-learning analysis back to the original application. A retailer may need interpretable customer segments, not simply a mathematical grouping.

Interpretation and Usefulness

A discovered pattern is not automatically useful just because an algorithm found it. The result should be considered in terms of:

- Relevance to the original problem
- Understandability, so the result actually makes sense
- Stability, so the same pattern keeps appearing
- Practical usefulness, so it supports real decisions
- Consistency with domain knowledge

Example: A customer group may be mathematically distinct but too small or too unclear to support a useful business decision. Discovery must be followed by interpretation and evaluation.

Evaluation Without Predefined Targets

In supervised learning, predictions can be compared with known target values. In unsupervised learning, such target answers are usually unavailable.

1. **Structure**
   Does the result show clear and meaningful organization?
2. **Consistency**
   Does the pattern remain reasonably stable?
3. **Interpretation**
   Can the result be explained in the application domain?
4. **Usefulness**
   Does it help answer the original problem?
5. **Validation**
   Can experts or later analysis support the discovery?

A strong unsupervised result should be both data-supported and meaningful in context — a pattern that only satisfies one of these two conditions is not yet a reliable discovery.

Example: Customer Segmentation

Complete Unsupervised Problem Formulation

Element	Formulation
Observation	One customer
Features	Spending, frequency, recency, product behavior
Target	No predefined segment
Objective	Discover meaningful customer groups
Output	Customer segments
Interpretation	Explain how the groups differ

This converts a broad business statement — “Understand our customers” — into a specific unsupervised-learning problem. The formulation should be clear before selecting a particular learning method.

Market Segmentation

Definition: **Market segmentation divides a large customer population into groups that show meaningful similarities in characteristics or behavior.**

Business motivation

- Customers often have different needs and spending patterns
- A single strategy may not suit the entire customer base
- Existing categories may be too broad or outdated
- New customer patterns may not be known beforehand
- Large populations are difficult to analyze manually

Objective: **Discover useful customer groups from available customer data — groups that the business can act on, not just groups that are mathematically distinct.**

An e-commerce company wants to understand how its customers differ

Business Goal and Customer Data

Available information

- Purchase frequency, showing how often the customer buys
- Total spending, showing the overall amount spent
- Average order value, showing the typical purchase size
- Recent activity, showing the customer's latest behavior
- Product preferences, showing which items are favored

The business does not provide predefined segment labels, so the goal is not to predict an existing customer type. It is to discover customer groups that show meaningful differences in behavior.

The value of the result depends on whether the discovered groups help the business understand or serve customers better.

Customer Representation

Customer A

- High spending
- Frequent purchases
- Recent activity, showing the customer's latest behavior
- Broad product interest

Customer B

- Low spending
- Occasional purchases
- Long gap since last purchase
- Narrow product interest

A good representation captures the behavior relevant to the segmentation objective. Including unrelated variables can distort the comparison and produce less useful groups.

Suppose the analysis produces several groups

Discovering and Interpreting Customer Groups

1

Group 1

Frequent, high-value customers

2

Group 2

Occasional, moderate-value customers

3

Group 3

Low-engagement customers

The grouping comes from the data. The descriptive names are added through interpretation. This matters because a learning method can discover a group, but it does not automatically know why that group matters to the business.

Value and Limitations of Segmentation

Possible value

- Target communication more effectively
- Understand different customer behaviors
- Support personalized offers
- Improve retention strategies
- Allocate marketing resources more carefully

Important limitations

- Results depend on selected features
- Different representations can produce different groups
- Very small groups may have little practical value
- A mathematical group may not have a useful business meaning

A useful segmentation should be interpretable, stable, and relevant to the business objective.

Customer Behavior Analysis

Definition: **Customer behavior analysis studies patterns in how users interact, purchase, search, or engage with a system. It is broader than simply dividing customers into groups.**

Possible behavioral information

- Purchase timing, showing when customers tend to buy
- Visit frequency, showing how often the customer returns
- Search activity, showing what customers are looking for
- Product preferences, showing which items are favored
- Response to promotions, showing reaction to discounts offered
- Changes in engagement, showing rising or falling interest

Objective: **Discover meaningful behavior patterns that are not explicitly labeled in the data — patterns of activity rather than a single fixed customer type.**

Behavioral Patterns and Signals

Behavior can contain repeated signals that are useful for discovery. For example, some users may:

- Purchase mainly after promotions
- Visit the platform frequently but spend little
- Make repeated purchases of similar products
- Become less active after a period of high engagement
- Show seasonal changes in their activity

The important point is that behavior can reveal patterns of activity, not just fixed customer categories.

Behavior Change Over Time

High Activity → Reduced Activity → Occasional Activity → Inactive

A dataset may contain no label such as “Declining customer engagement,” yet the sequence of activity may reveal a meaningful pattern. This makes time and behavioral change important when analyzing customer data.

Segmentation vs. Behavior Analysis

Market Segmentation	Customer Behavior Analysis
Focuses on groups	Focuses on behavior patterns
Often uses customer profiles	Can examine activity and change
Produces customer segments	Produces behavioral insights
Useful for understanding groups	Useful for understanding how behavior occurs
Often based on a snapshot	Can include behavior over time

Both may use the same customer data. The difference is the question being investigated and the type of knowledge required.

Streaming Platform — records viewing duration, genres watched, search activity, session frequency, viewing time

Complete Customer Behavior Case

Possible findings

- Frequent short sessions, suggesting quick repeated visits
- Long viewing sessions, suggesting deep sustained engagement
- Strong preference for particular genres
- Gradually declining engagement, suggesting fading interest over time

The goal is not necessarily to create customer segments. The analysis may instead reveal behavioral patterns and changes in engagement.

Anomaly Detection

Definition: **Anomaly detection identifies observations that differ significantly from a dominant or expected pattern in the data.**

An anomaly may be interesting because it

- Differs strongly from normal behavior
- Appears rarely compared with most other observations
- Represents an unexpected combination
- Requires additional investigation before any conclusion is drawn
- May indicate a change, error, or unusual event

The purpose is usually to identify observations that deserve attention, not to immediately determine their cause.

Normal Behavior, Deviation, and Context

An observation should not be called unusual only because its value is large or small; what matters is how far it sits from the typical pattern for its own context.

Example	A ₹50,000 transaction may be unusual for a student account, but normal for a business account

Similarly, a high number of daily logins may be normal for one user but unusual for another. Anomaly detection depends on the reference pattern and context used for comparison — the same observation can be normal in one context and unusual in another.

Anomaly vs. Fraud

Anomaly	Fraud, where the activity is intentionally deceptive
An observation that differs from the expected or dominant pattern.	***Activity intended to deceive or cause unauthorized loss.***

An anomalous transaction could result from

- Fraud, where the activity is intentionally deceptive
- A genuine exceptional purchase
- A new customer behavior
- A data error, such as a system glitch
- Another rare but valid event

Anomaly detection identifies candidates for investigation; it does not automatically prove fraud.

Fraud Detection

Financial systems process very large numbers of transactions every day, which makes manual review of every case impossible and automated detection necessary.

Useful transaction information

- Transaction amount, showing how much was spent
- Time, showing when the transaction took place
- Location, showing where the transaction occurred
- Merchant, showing where the purchase was made
- Transaction frequency, showing how often transactions occur
- Customer history, showing the customer's past behavior

A transaction that differs strongly from established behavior may deserve investigation. Unsupervised approaches are useful when complete fraud labels are unavailable or when new forms of unusual behavior have not been seen in labeled historical data.

Why Fraud Is Difficult to Label

Fraud detection is difficult because the data itself keeps changing — what counts as normal behavior today may not describe tomorrow's new fraud strategy.

- Fraud events are relatively rare
- Historical labels may be incomplete
- New fraud strategies can appear
- Previously unseen patterns may have no labels
- Normal behavior differs between customers
- Millions of transactions make complete manual review unrealistic

This makes the discovery of unusual behavior valuable even when a complete set of fraud labels is unavailable.

Fraud Detection Case

Customer's normal range	₹1,500–₹5,000 transactions, mainly from familiar locations
New transaction	₹48,000 from a new location

Interpretation: **The transaction is significantly different from the customer's established behavior, which makes it a reasonable candidate for review even before any cause is known.**

Appropriate system response: **Flag for investigation — rather than automatically declare fraud. The anomaly provides a reason to investigate further.**

Pattern Discovery in Biological Data

Biological datasets can contain a very large number of measurements across many samples, often far more variables than there are samples to compare.

Examples

- Gene-expression measurements, showing which genes are active
- Protein measurements, showing levels of specific proteins
- Molecular characteristics, describing the sample at a molecular level
- Cell-level measurements, describing individual cell properties
- Biological markers, signaling a specific biological condition

Discovery goals: **Researchers may look for similar biological profiles, potential subgroups, distinct molecular patterns, unusual samples, and relationships among biological variables. The discovered structure can support further scientific investigation.**

Gene-Expression and Disease-Subtype Patterns

Suppose researchers measure the expression of thousands of genes across many biological samples. The samples may not initially have clearly defined subtypes.

Pattern discovery can help identify samples with similar expression profiles and may reveal candidate biological subgroups or disease-related patterns.

Important distinction: **The computational analysis identifies candidate structure. Biological experts must still determine whether the discovered pattern has a meaningful scientific interpretation. This keeps pattern discovery separate from scientific validation.**

Pattern Discovery in Social Data

Social platforms generate large datasets from connections and interactions between users, and structure in this data comes from how people relate to one another rather than from any single profile.

- User connections, showing who is linked to whom
- Messages and replies, showing direct communication between users
- Shared content, showing what users post or share
- Interaction frequency, showing how often users engage
- Likes and engagement, showing how users respond to content
- Participation in groups or communities

Unlike a predefined customer category, social structures may emerge from the interaction data itself. Typical discovery goals include identifying communities, similar interaction patterns, and changes in social behavior.

Social Community and Interaction Patterns

A social network can contain groups of users who interact more frequently with one another than with the rest of the network.

Possible patterns

- Closely connected user communities
- Users with similar interaction behavior
- Highly connected individuals who link many users together
- Repeated interaction relationships between the same group of users
- Changes in community structure over time

The discovered structure still needs interpretation. A group may represent a shared interest, professional network, event, friendship pattern, or another type of relationship.

Use-Case Comparison: Five Real-World Settings

Application	Main Observation	Discovery Goal
Market segmentation	Customer	Find customer groups
Customer behavior	User activity	Find behavior patterns
Anomaly detection	Transaction/event	Find unusual behavior
Biological data	Sample	Find biological patterns
Social data	User/network	Find communities or interaction patterns

The application changes, but the basic process remains: Represent the data → define the objective → discover structure → interpret the result. The important difference is the kind of knowledge required by each domain.

Business vs. Scientific Discovery

Business Applications

- Which customers behave similarly?
- Which transactions need investigation?
- Which groups may require different strategies?

Scientific Applications

- Find previously unknown biological patterns
- Identify possible subgroups
- Generate new research hypotheses
- Understand complex interaction structures

The discovery process may be similar, but the meaning and validation of the result are different.

Identify the discovery objective

Use-Case Knowledge Check

Match from: **Relationship Discovery · Group Discovery · Anomaly Discovery · Community Discovery**

1. Group customers with similar purchasing behavior.

2. Identify transactions that differ strongly from normal behavior.

3. Find recurring relationships between items or events.

4. Identify groups of users with dense social interactions.

Answer key: **1 → Group Discovery 2 → Anomaly Discovery 3 → Relationship Discovery 4 → Community / Group Discovery**

Distance and Similarity Metrics

Many unsupervised-learning problems require comparing observations. For example:

- Which customers are most similar?
- Which transaction is farthest from normal behavior?
- Which documents discuss similar content?
- Which biological samples have similar profiles?

A machine needs a measurable way to make these comparisons. This is the role of distance and similarity metrics.

Distance vs. Similarity

Distance

Measures how separated or different two observations are.

Smaller distance → closer. Larger distance → farther apart.

Similarity

Measures how closely two observations resemble one another.

Higher similarity → greater resemblance.

Both concepts help define relationships between observations, but different metrics capture different meanings of “close” or “similar.”

Why Similarity Must Be Measured

"These customers seem similar."

A machine needs a measurable rule to support that judgment. The chosen measure can influence:

- Which observations are considered neighbors
- Which customers appear similar
- Which observations look unusual
- How groups are formed
- How documents are compared

Distance or similarity is not just a mathematical detail — it affects what the system considers related.

Different Meanings of “Similar”

Two observations can be considered similar in different ways, and the choice of metric later depends entirely on which of these meanings actually matters for the problem.

1. **Geometric similarity**
   They are close in numerical feature space

2. **Coordinate-wise similarity**
   Their individual feature differences are small

3. **Directional similarity**
   Their feature vectors point in a similar direction

These different interpretations lead to different metrics. Understanding the meaning of similarity should come before choosing a particular metric.

Euclidean Distance

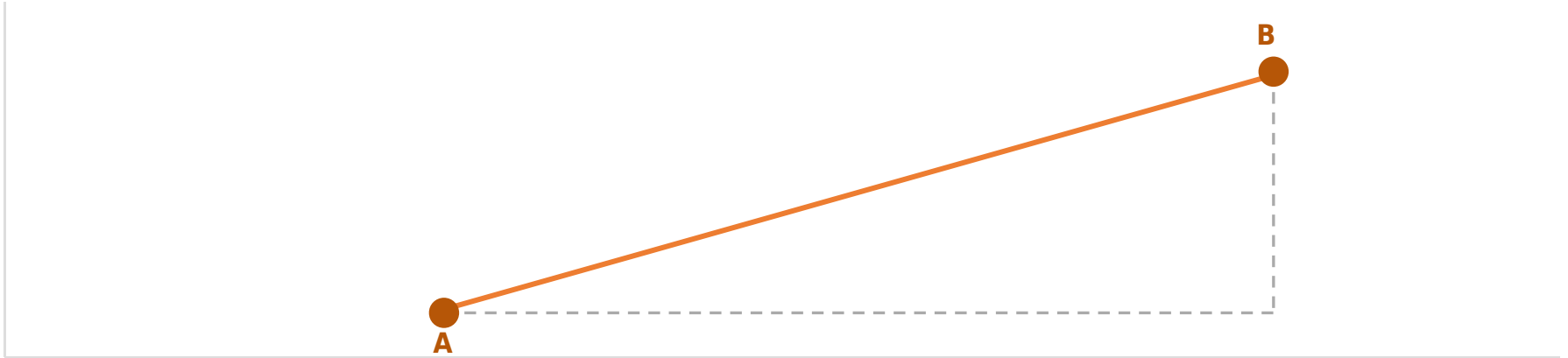
Definition: **Euclidean distance measures the straight-line separation between two observations in a feature space.**

- Natural geometric interpretation that matches everyday intuition
- Common for numerical features
- Measures overall separation between two observations directly
- Symmetric between two observations
- Sensitive to feature scale
- Useful when geometric closeness is meaningful

The simplest intuition is the direct distance between two points — the same way a ruler would measure the gap between them on paper.

Euclidean Geometric Interpretation

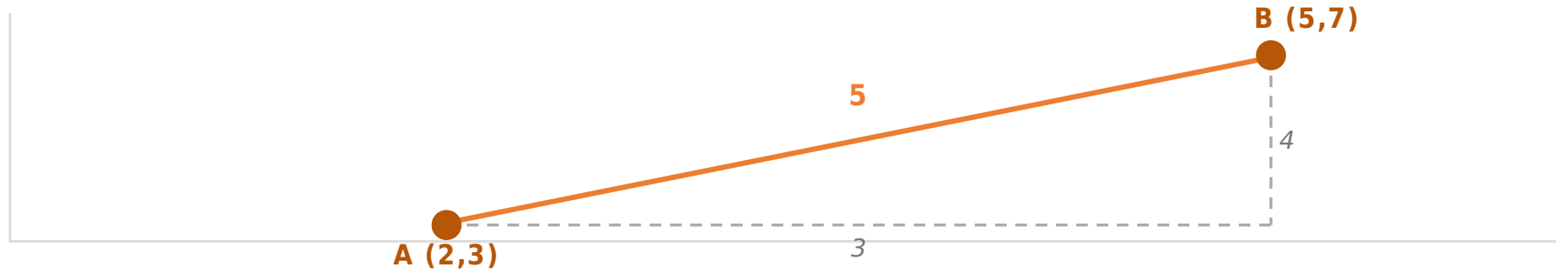
Imagine two observations plotted as points on a 2D graph.



Euclidean distance represents the shortest straight-line path between the two points — often described as the “as the crow flies” distance. The interpretation is useful when the coordinates themselves represent meaningful numerical dimensions.

Euclidean Distance: Worked Calculation

A	(2, 3)
B	(5, 7)
Differences	First dimension: 5 − 2 = 3. Second dimension: 7 − 3 = 4



These differences form a 3–4–5 triangle. Euclidean distance = 5 units. The example shows how differences across multiple features are combined into one geometric distance.

Suppose customers are represented using Purchase Frequency and Average Spending

Euclidean Distance in Feature Space

Each customer becomes a point in a feature space. Customers with similar values appear closer; customers with substantially different values appear farther apart.

This makes Euclidean distance intuitive for numerical customer data, provided that the features are represented on suitable scales.

Important: Distance is meaningful only when the underlying feature space is meaningful — comparing points on arbitrary or poorly chosen axes produces numbers that look precise but mean little.

Euclidean Distance: Properties

- It is always non-negative
- The distance between an observation and itself is zero
- The distance is symmetric between two observations
- Greater geometric separation produces a larger distance
- It extends naturally from 2D to higher-dimensional numerical data
- It provides a familiar geometric interpretation

These properties make Euclidean distance widely useful in numerical data analysis, since they guarantee the measure behaves consistently no matter which two observations are compared.

Euclidean Limitations and Scale Sensitivity

Euclidean distance can be strongly influenced by feature scale, so two features measured in very different units can silently distort the comparison.

Height	Measured in centimeters
Weight	Measured in kilograms

The numerical ranges and units are different. A large-scale feature can contribute more strongly to the distance even when the application does not intend that feature to dominate.

Therefore: **Before using Euclidean distance, feature scaling and the meaning of each variable should be considered carefully.**

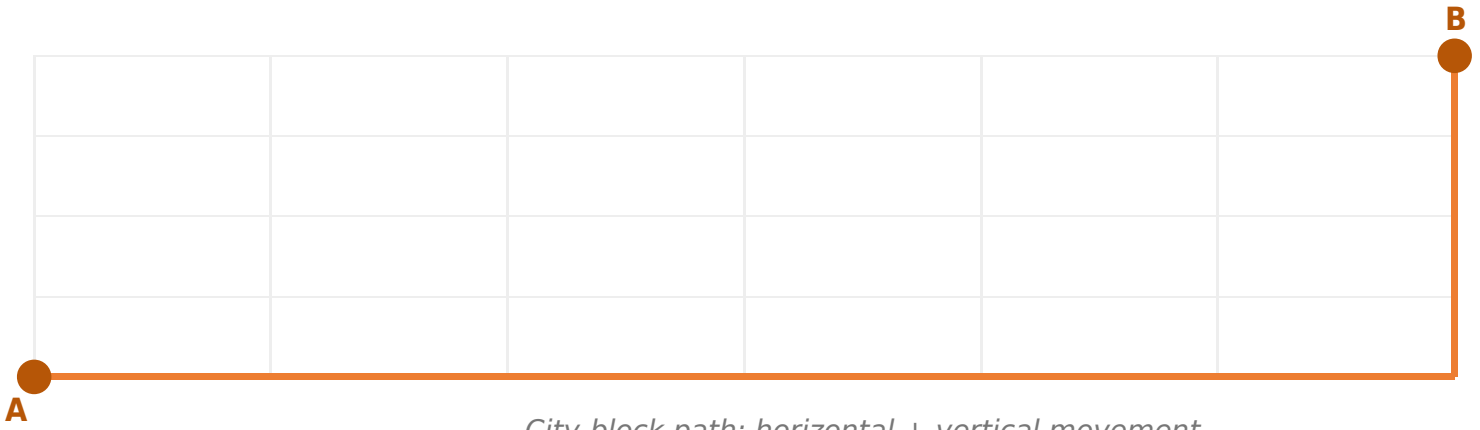
Manhattan Distance

Definition: **Manhattan distance measures the total absolute difference between two observations across their individual dimensions. Unlike Euclidean distance, it does not use the shortest diagonal path.**

- Compares each dimension separately rather than combining them
- Adds up the absolute differences across all dimensions
- Easy to interpret because it mirrors physical movement
- Useful when coordinate-wise change is meaningful
- Sensitive to feature scaling
- Produces a different notion of distance from Euclidean

City-Block Interpretation

Imagine moving through a city where streets form a rectangular grid. A person cannot travel directly through buildings — the route follows the streets: horizontal movement + vertical movement.



City-block path: horizontal + vertical movement

Manhattan distance follows the same idea in feature space. It measures the total change needed across the individual dimensions rather than the shortest direct path.

Manhattan Distance: Worked Calculation

A	(2, 3)
B	(5, 7)
Differences	$\|5 - 2\| = 3$. $\|7 - 3\| = 4$
Manhattan distance	3 + 4 = 7 units

For the same observations: Euclidean distance = 5, Manhattan distance = 7. The different values come from using different definitions of distance.

Euclidean vs. Manhattan

Aspect	Euclidean	Manhattan
Path	Straight line	Coordinate-wise
Interpretation	Geometric separation	Grid-like movement
Calculation	Combines differences geometrically	Adds up the absolute differences across all dimensions
Scale sensitivity	Yes	Yes
Suitable when	Overall geometry matters	Dimension-wise differences matter

Neither metric is universally better. The appropriate choice depends on how separation should be interpreted for the problem.

Manhattan: Properties and Limitations

Properties

- Non-negative distance, since separation cannot go below zero
- Distance from a point to itself is always zero
- Symmetric between two observations either way
- Easy coordinate-wise interpretation that matches everyday movement
- Useful when individual feature differences are meaningful

Limitations

- Still affected by feature scale
- Irrelevant features can distort the result
- May not represent similarity well for directional data
- The choice should reflect the meaning of each feature

Manhattan changes the definition of closeness; it does not automatically make the comparison better.

Cosine Similarity

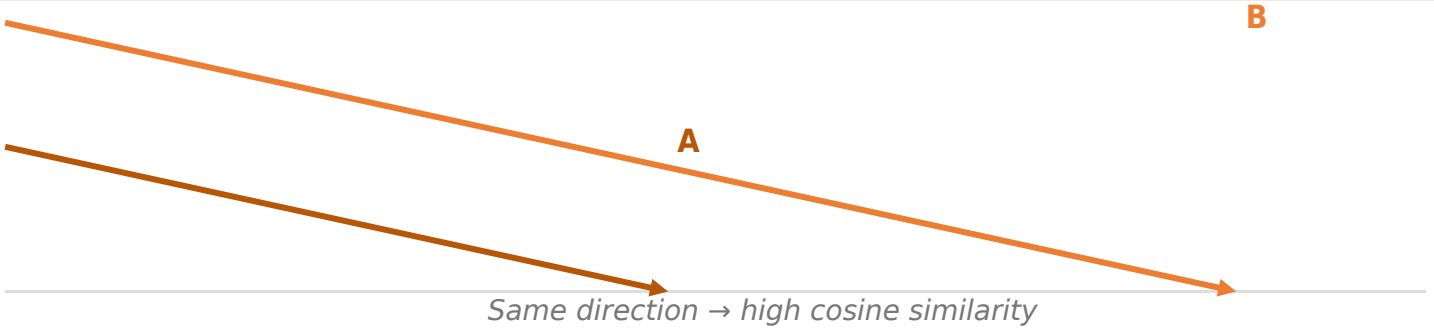
Definition: **Cosine similarity measures how closely two vectors point in the same direction, rather than mainly comparing their size.**

- Compares the orientation of two vectors
- Useful when feature proportions matter
- Less affected by overall magnitude
- Commonly used with text and sparse vectors
- Higher similarity means stronger directional agreement

Cosine similarity is different from Euclidean and Manhattan distance: it focuses on direction rather than geometric separation.

Direction vs. Magnitude

A	(1, 2)
B	(2, 4)
Observation	Vector B is twice the size of vector A, but both point in exactly the same direction



Magnitude is different; direction is the same. Their cosine similarity is 1. This is useful when two observations should be considered similar because their feature pattern is similar, even when one has much larger overall values.

Cosine Similarity: Worked Calculation

For two vectors A and B, cosine similarity compares their dot product with the product of their magnitudes.

A	(1, 2)
B	(2, 4)
Relationship	Every component of B is twice the corresponding component of A — same direction
Cosine Similarity	1

Interpretation: **1 → same direction. 0 → perpendicular direction. Lower values → weaker directional similarity. The important idea is not only the value, but what it means about the orientation of the vectors.**

Cosine Similarity for Text Vectors

Term	Document A	Document B
cloud	5	2
server	4	2
storage	3	1
network	2	1

The documents have different total counts, but their term patterns are similar. Cosine similarity can identify this directional similarity without treating the longer document as completely different simply because it contains more words.

TF-IDF Document Example

TF-IDF converts documents into numerical vectors based on the importance of terms. Consider two news articles, both mainly about cloud computing — their lengths may differ, but the important terms can have similar relative patterns.

Cosine similarity can compare those vectors to estimate how similar the documents are, based on which terms appear together rather than on how long each document is.

Common uses

- Document comparison, checking how similar two texts are
- Search systems, ranking results by relevance to a query
- Information retrieval, finding documents that match a need
- News recommendation, suggesting articles similar to past reads
- Similar-content detection, finding duplicate or near-duplicate text

Cosine Similarity: Properties and Limitations

Properties

- Focuses on vector direction
- Useful for high-dimensional representations
- Works well with sparse feature vectors
- Less sensitive to overall magnitude
- Particularly useful for text and document representations

Limitations

- The meaning of direction must make sense for the data
- It does not directly represent geometric distance
- Poor feature representation can still produce misleading similarity
- Not automatically the best choice for every numerical dataset

Euclidean vs. Manhattan vs. Cosine

Metric	Main Idea	Typical Use
Euclidean	Straight-line separation	Numerical geometric data
Manhattan	Sum of coordinate differences	Grid-like or coordinate-wise data
Cosine	Directional similarity	Text and sparse vectors

Practical distinction: **Euclidean: how far apart are the points? Manhattan: how much coordinate-wise change separates them? Cosine: how similarly are the vectors oriented? The choice should follow the meaning of similarity in the dataset.**

Metric Choice and Data Representation

Customer profile: Age + income + spending	Geometric numerical differences may be meaningful
Product preference vector	The direction or pattern of preferences may matter more
Document vector: Word / TF-IDF features	Directional similarity is often useful

Metric selection should begin with the question: how have the observations been represented, and what should “similar” mean?

Feature Scaling and Distance

Height	150–200 cm
Annual income	₹2,00,000–₹20,00,000

A numerical distance may be dominated by income because its values are much larger. This can make the comparison reflect scale rather than the intended importance of the features.

Common preparation approaches

- Standardization, rescaling features onto a common range
- Normalization, adjusting values onto a comparable scale
- Reviewing feature units so scales stay comparable
- Removing irrelevant variables that add no real value

Suppose customer similarity is measured using Annual income + monthly purchase count

What Happens Without Scaling?

Without scaling, income may contribute much more to the numerical distance simply because of its larger values. A customer with similar purchasing behavior could still appear far away because of an income difference.

Consequence: **The algorithm may discover groups based mainly on the large-scale feature. A mathematically correct distance can still produce an unsuitable result when the feature scales are poorly handled.**

High-Dimensional Distance

When the number of features becomes very large, distance becomes harder to interpret, because each additional dimension adds its own noise to the overall comparison.

- Many dimensions contribute to each comparison
- Irrelevant features can add noise
- Observations may appear similarly distant
- Meaningful differences can become harder to isolate
- Visualization becomes difficult beyond just a few dimensions

This issue is related to the curse of dimensionality. Increasing the number of features does not always make similarity measurement more informative.

Poor Metric Choice and Misleading Results

Groups created using	Age + Income
Real business objective based on	Purchased Products + Purchase Frequency

The algorithm may find clearly separated groups, but those groups may not describe the business behavior the company actually cares about.

Lesson: **A good metric cannot fix a poor representation or a poorly defined objective. Metric choice must match the data and the purpose of the analysis.**

Which Metric Fits the Dataset?

Match from: **Manhattan Distance · Cosine Similarity · Euclidean Distance**

A Two numerical features represent physical measurements, and geometric separation is meaningful.

B Observations are represented as text vectors, and document length should have less influence than word pattern.

C The total change across individual dimensions is the meaningful measure of difference.

Answer key: **A → Euclidean B → Cosine Similarity C → Manhattan**

Decision rule: **Choose the metric according to: Data type + representation + meaning of similarity + analytical objective.**

Unit I: Complete Knowledge Map

Category	Sequence
Learning Paradigms	Supervised → Unsupervised → Semi-Supervised
Unsupervised Problem	Observation → Features → Objective → Output → Interpretation
Real-World Applications	Market Segmentation → Customer Behavior → Anomaly/Fraud → Biological → Social
Similarity	Distance → Euclidean → Manhattan → Cosine → Metric Choice

Final integration: **A successful unsupervised-learning problem requires more than an algorithm. It requires a clear objective, suitable representation, meaningful comparison, and careful interpretation of the discovered structure.**